

(The author used pins 2 and 13.)

Next, set the display of the display driver GC9A01 in the code. Setting the board and pins for the display is shown in Fig. 3.

Now we need to set the Wi-Fi SSID and password for the NTP server so that the watch can connect to Wi-Fi of our phone or the home Wi-Fi to fetch the time from NTP client-server. So, we create the setup function where we initialise the I2C communication with the MAX30100 sensor, the SPI communication with the LCD display, and the NTP server for the clock. Wi-Fi setting for the NTP server is shown in Fig. 4.

Creating GUI for watch

Create a loop function where the sensors and time data can be updated in real time. Then create an interface and look for the smart-

watch to display the health data, date, and time data in real time.

You can code and make any style of GUI of your choice for the watch. The author used the circular mark with hands for minutes and hours with a circular arc that decreases with every second to show the time in seconds. He used an arc and circle with text to display the SpO2 value and a round rectangle for displaying the heartbeat data. He used text to display the current day, month, and other time-related data.

After preparing the code upload the code to the ESP12F chip by selecting the right board and port number. The chip needs to be set to boot mode and connected to the FTDI programmer for that. (To know how to upload code to the ESP12F chip, refer programming part of my article on IoT health sticker published in December issue.)



Fig. 8: Smart watch inside 3D printed case



Fig. 10: Smart watch showing time and health data

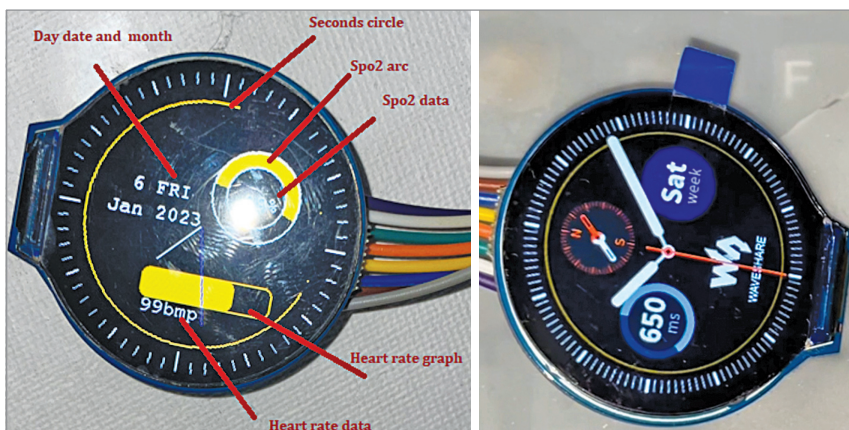


Fig. 9: GUI of author-designed watch (LHS) and another GUI style you can try

Construction and testing

Before assembly, refer Fig. 5 through Fig. 8. Place ESP12F behind the display and Max3011 sensor carefully such that their terminals and connection points do not touch and short. Get the watch's circular case and fit the components inside it. If you wish, you can design your own watch case and 3D print it. You can download the Gerber file and PCB design to get the watch ready.

Create an interface for the smart watch to display the health data, date, and time data in real time. You can code and make any style of GUI of your choice.

Fig. 5 shows PCB design for the watch. Soldering of ESP12F chip with display and placing the ESP chip behind the watch is shown in Fig. 6. Smart watch 3D case design is shown in Fig. 7. Fig. 8 shows the smart watch inside 3D printed case.

After preparing the code, upload the code to ESP12F chip by selecting the right board and port number. The chip needs to be set to boot mode and connected to the FTDI programmer. GUI of the watch designed by the author is shown in Fig. 9.

To test, power the watch with a 3V battery and wait till it connects to Wi-Fi and fetches current day, date, and time from the NTP server. It is now ready to start showing the health data besides time. Final look of the programmable smart watch with fitness tracking is shown in Fig. 10.

Note. This is the first version of the smart watch design. An advanced design with additional functions like a real-time Wi-Fi analyzer, web health GUI, and many more is getting ready for publication in a coming issue. **EFY**

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